

Organization of Somatosensory Cortical Areas in the Naked Mole-Rat (*Heterocephalus glaber*)

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ABSTRACT

Multiunit electrophysiology was combined with histological analysis of cortical sections to investigate the organization of somatosensory areas in the naked mole-rat. We provide new details for the organization of primary somatosensory cortex (S1) and identify cortical modules and barrels that correspond to the representations of different body parts. In addition, details of the location and organization of secondary somatosensory cortex (S2) are reported, and evidence for a third somatosensory representation, likely the parietal ventral area (PV), is provided and discussed. S1 contained a complete and systematic representation of the contralateral body surface and oral structures. The orientation of S1 was inverted, with the lower body represented medially and the face and oral structures located rostrally. The S2 representation was found in caudolateral cortex forming a mirror image of S1. The two areas were joined at the representation of the vibrissae and snout, so that the orientation of S2 formed an upright representation of the body in cortex. Receptive fields for S2 were consistently larger than those in S1. Evidence for the presumptive parietal ventral area, lateral to S2, suggests that this area may be an inverted mirror image of S2. By aligning the electrophysiological maps of body representations with cytochrome oxidase-reacted cortical sections we were able to identify modules related to the buccal pad, chin, vibrissae, forelimb, hindlimb, trunk, tongue, lower incisors, and upper incisors. The orofacial modules in lateral cortex resemble similar modules reported to relate to oral structures previously described in the laboratory rat, owl monkey, and squirrel monkey. *J. Comp. Neurol.* 495:434–452, 2006.

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Naked mole-rats are remarkable rodents with many unusual traits. They live in large subterranean colonies of 80 or more individuals and have only a single reproductively active female (Jarvis, 1991; Honeycutt et al., 1991). This eusocial colony structure includes separate castes—including a queen, many workers, and occasional disperser morphs—thus closely resembling the organization of social insect colonies such as ants and termites. Naked mole-rats also have extraordinarily long lives for rodents, with captive animals living over 20 years (O'Connor et al., 2002).

Our interest in naked mole-rats stems from the sensory specializations that allow them to navigate their dark underground tunnels. As might be expected, naked mole-rats have reduced eyes with sparse visual projections to the lateral geniculate nucleus and superior colliculus, al-

though strong projections to the superchiasmatic nucleus remain (Dengler-Crish et al., 2004, and see Bronchti et al., 1991; Cooper et al., 1993a,b; Negroni et al., 2003, for other mole-rat species). They have few retinal ganglion cells compared to sighted rodents (Cernuda-Cernuda et al.,

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Fig. 1. The naked mole-rat (*Heterocephalus glaber*). This subterranean mammal is virtually hairless with only a sparse collection of body and facial vibrissae. Note the large, extra-oral incisors and reduced eyes and ears. Scale bar = 1 cm.

2002) with a disorganized retinal architecture that prevents functional image formation (Mills and Catania, 2004; Nikitina et al., 2004).

Not surprisingly, mole-rats depend heavily on touch to navigate their environment. Although virtually hairless, they have unique “body vibrissae” in rows along the trunk that act as touch sensors for navigation within tunnels and for accurately orienting their head in the direction of novel stimuli (Crish et al., 2003). They also have facial whiskers more typical of other rodents (Fig. 1).

But perhaps the most obvious features of their external anatomy are the large incisors that protrude beyond the oral cavity. The mandibles connect at their anterior extent by a flexible mandibular symphysis, allowing the lower incisors to move independently and providing a greater range of movements and dexterity than is typical of most mammalian teeth (Catania and Remple, 2002). Mole-rats use their teeth for digging extensive tunnel networks, carrying objects and young, chewing, and for investigating their surroundings (by tapping their incisors against objects of interest).

All of these somatosensory specializations appear to compensate for reduced eyes and ears, and demonstrate the effective way that this animal has adapted to its environment. As might be expected, the peripheral sensory specializations described above are reflected in an equally specialized and expanded somatosensory cortex. In a previous brief report we described the overall layout of primary somatosensory cortex (S1) in naked mole-rats and compared the sizes of various body part representations in mole-rats with corresponding representations in laboratory rats (Catania and Remple, 2002). A striking result was the discovery of a greatly expanded representation of the front teeth in mole-rats, which occupied 31%

of their large somatosensory cortex. However, a number of questions remained regarding the organization of naked mole-rat cortex. For example, there was a series of histologically visible modules and subdivisions in the cortex of mole-rats, but it was not clear how these modules related to the representations of different sensory areas and body parts. Similar modules in the cortex of mice (Woolsey and Van der Loos, 1970), rats (Welker, 1971, 1976), and primates (Jain et al., 2001) have provided important general insights into how cortex is subdivided across mammals (Catania, 2002) and also provided landmarks for a range of experimental procedures. In this study we provide evidence for how these modules in mole-rat cortex relate to different sensory representations.

In addition, the organization and position of S2 remained uncertain in mole-rats and it was unclear whether additional somatosensory areas exist. A number of investigations suggest that additional cortical subdivisions may be common to most mammals (e.g., Krubitzer et al., 1986, 1995a; Krubitzer and Kaas, 1990; Disbrow et al., 2000; Slutsky et al., 2000). Here we investigate these regions in mole-rats and provide additional information useful for determining the ancestral organizational plan for somatosensory areas in rodents and mammals generally.

Finally, naked mole-rats depend more heavily on somatosensation than most species and by examining the details of their cortical organization we provide information about how the neocortex evolves in concert with a changing sensory periphery, e.g., how cortical space allocation correlates with expansions and reductions of different sensory systems. These questions are approached by examining and comparing the anatomy and electrophysiology of somatosensory areas in the neocortex of naked mole-rats.

MATERIALS AND METHODS

Five adult animals were used to determine the organization of the somatotopic cortical representations in the naked mole-rat. All animal procedures followed National Institutes of Health guidelines and were performed according to the standards set by the Animal Welfare Act and the Vanderbilt University Institutional Animal Care and Use Committee. The animals were anesthetized with 15% urethane diluted in distilled water at a dosage of 1.5 g/kg. Body temperature was maintained with a heating pad and hot water bottles. A headpost was attached to the right side of the skull using dental cement and a craniotomy was performed to expose the left cortex. The cortex was covered with silicon and a digital photograph of the cortical surface was taken to mark penetration sites during the recording session.

Multiunit physiology recordings were performed using low impedance tungsten electrodes (1.0 M Ω at 1,000 Hz) at cortical depths ranging from ~400–500 μ m. To prevent potential bias in receptive field evaluation, one investigator was responsible for positioning the electrode in each cortical site while a second investigator identified receptive fields. Soft probes and calibrated von Frey hairs were used to identify the receptive fields for each penetration. The boundary of the receptive field was defined as the skin surface (or tooth) where light contact produced a detectable increase in neural activity as visualized with an oscilloscope combined with auditory monitoring through a speaker. Each receptive field was drawn on a schematic of

TABLE 1. S1 (top) and S2 (bottom) Receptive Field Area Averages

S1 receptive field area averages for each recording case¹

Body part	Case 1 (01–15), mm ²	Case 2 (01–13), mm ²	Case 3 (01–11), mm ²	Case 4 (01–16), mm ²	Case 5 (01–14), mm ²	Compiled average, mm ²
Head/Vibrissae*	14.48	5.40	3.40	10.11	10.80	8.84
Buccal Pad	2.58	2.60	2.51	3.14	3.16	2.80
Chin	8.31	4.77	4.70	8.11	3.84	5.95
Trunk/Tail	198.93	58.10	133.49	106.54	191.65	137.74
Forepaw	32.20	26.56	37.48	64.09	25.72	37.21
Hindpaw	43.69	45.82	38.47	37.96	32.49	39.69

¹Data are expressed as the mean of five randomly selected receptive field areas within each S1 body representation category.

S2 receptive field area averages for each recording case

Body part	Case 1 (01–15), mm ²	Case 2 (01–13), mm ²	Case 3 (01–11), mm ²	Case 4 (01–16), mm ²	Case 5 (01–14), mm ²	Compiled average ² , mm ²
Head/Vibrissae ³	81.68(n = 5)	8.20(n = 5)	11.98(n = 1)	11.02(n = 1)	99.82(n = 2)	48.00
Buccal Pad	8.46(n = 2)	5.12(n = 3)				6.46
Chin					21.27(n = 1)	21.27
Trunk/Tail	697.97(n = 16)	559.73(n = 5)			409.68(n = 4)	624.20
Forepaw	120.46(n = 6)	91.14(n = 2)		66.34(n = 1)		107.93
Hindpaw	192.46(n = 2)	108.86(n = 3)			200.33(n = 1)	152.01

²Compiled average is calculated as a weighted mean.³Significant difference between S1 and S2 Head/Vibrissae receptive field averages ($p = 0.043$).

a mole-rat body. Selected electrode penetrations were marked with microlesions (10 μ A while retracting from a depth of 1,000 μ m at a speed of 70 μ m/s). At the end of the recording session, each animal was given an overdose of sodium pentobarbital (100 mg/kg) and perfused with 0.9% saline followed by 4% paraformaldehyde. The brain was removed and the cortical hemispheres were separated from underlying white matter and subcortical tissue using a small, smooth spatula (Robox RS-8820, Gaithersburg, MD) and flattened between glass slides. The cortices were immersed in 30% sucrose in phosphate buffer for 12 hours and then the flattened cortex was frozen to a flat ice block on the stage of a freezing microtome and tangential sections were cut at 60 μ m. Cortical sections were processed for the metabolic enzyme cytochrome oxidase (CO), which often reveals sensory areas (Wong-Riley and Carroll, 1984). Sections were photographed using a 35-mm digital camera mounted to a microscope. Imported images were adjusted for brightness and contrast using PhotoShop 7.0 (Adobe, San Jose, CA). Illustrator 10 (Adobe) was used to align the recording sites marked on brain photograph with images of the cortical sections using microlesions as landmarks. Outlines were drawn around groups of penetration sites related to each body part or orofacial region. Receptive field progressions were drawn on schematics of the naked mole-rat.

The data for the S1 receptive field area averages in Table 1 were generated by randomly selecting five electrode penetrations marked on the brain photograph for each body part category. The numerical system we used to record the microelectrode penetrations on an enlarged image of the exposed brain did not contain information related to the receptive field schematics, allowing the researcher to select sites while blind to their corresponding receptive fields. To determine the receptive field area averages of S2 recording cases, all penetration sites recorded in S2 were included. The receptive field drawings were scanned into the computer and imported into Illustrator 10. The area of each receptive field was calculated using CADTools3 (Hot Door, Laguna Beach, CA), which was scaled so that the mole-rat schematics were normalized to

a 12-cm mole-rat body length (the average length of an adult nonreproductive mole-rat in our colonies). The mean of the receptive field areas for each recording case (categorized by body part) are displayed along with compiled averages for each body part across the five cases. A Wilcoxon Signed-Ranks test was performed on the head/vibrissae dataset (the only category which contributed data from all five cases across both S1 and S2) to test for statistical significance.

RESULTS

The results of this study provide a description of the topography, orientation, and histological correlates of the cortical somatosensory areas in the naked mole-rat. We identified cytochrome oxidase-dense modules and barrels that relate to the S1 body representation and orofacial structures. We also report the organization and topography of S2 and provide tentative evidence for the parietal ventral area (PV).

Histological appearance of layer IV cortex

Figure 2 shows the gross morphology of the naked mole-rat brain with an overview of the size and location of cortical areas identified in this investigation. As expected for this small mammal, the cortex was lissencephalic and divided into piriform cortex and neocortex by the rhinal sulcus at the approximate level of the zygomatic arch. Much of the caudal half of cortex was occupied by the somatosensory areas, with little room remaining for visual cortex. The face and body representations in S1 were shifted in the medial direction compared to other rodents, which may be a result of the greatly expanded incisor representations in rostralateral cortex. In addition to the medial shift in the overall position of the body representation, S1 was considerably expanded in the caudal direction.

The histological appearance of layer 4 cortex processed for cytochrome oxidase is shown in Figure 3, distinguishing cortical regions with high metabolic activity (Wong-Riley et al., 1978; Wong-Riley and Welt, 1980). Careful

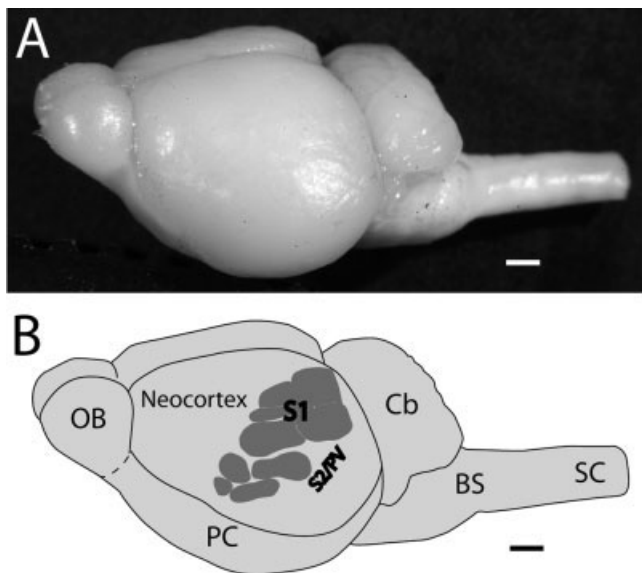


Fig. 2. The brain of the naked mole-rat. Photograph (A) and drawing (B) of brain with labels, showing the relative size of the olfactory bulbs (OB), neocortex, piriform cortex (PC), cerebellum (Cb), brainstem (BS), and spinal cord (SC). Scale bars = 1 mm.

inspection of the caudal and medial area of the cortex reveals a cytochrome oxidase pattern with distinct parallels in overall form to patterns described in other rodents (e.g., Woolsey and Van der Loos, 1970; Welker, 1976; Dawson and Killackey, 1987). For example, an area of cytochrome oxidase (CO)-dense barrels is evident caudally, merging rostrally with a more uniform CO-dense region. Just medial to these areas, at least two more modules can be identified rostral to a uniformly dark region in more caudomedial cortex. Note, however, the medial position of these modules representing the body compared to similar modules in the somatosensory cortex of laboratory rats (e.g., Welker, 1976; Dawson and Killackey, 1987; Remple et al., 2003).

In addition to this histological reflection of the head, trunk, and limbs, multiple modules were found in anterolateral cortex related to the representation of oral structures in S1 (Figs. 3, 4). Similar but smaller modules have been identified in the somatosensory cortex of laboratory rats (Remple et al., 2003). The lower panel in Figure 3 shows how the different modules relate to various body parts and oral structures. Evidence for these assignments comes from our extensive electrophysiological recordings from cortex (Figs. 5–12). Note for example the correlation between the location of body parts (vibrissae, chin, forelimb, hindlimb, and trunk) in the electrophysiological map in Figure 5 (previously published in summary form; Catania and Remple, 2002) and the more recently identified CO-dense modules in Figure 3. Although this pattern is nearly identical, in outline, to the pattern of CO-dense areas in laboratory rodents (Dawson and Killackey, 1987), barrels are less obvious and more variable in mole-rat cortex. This may reflect the relatively smaller whiskers and buccal hairs on the face of the naked mole-rats compared to laboratory rats and mice.

The relationship between the modules in the rostralateral cortex and the oral structures was less obvious; however, in a number of cases we were able to pair microlesions with physiological recordings to relate receptive fields on the face to each cortical module. We identified two modules that corresponded to the lower incisor representation (LIM1 and LIM2), one module that corresponded to the upper incisor area (UIM), and one that corresponded to the tongue area (TM). The lower incisor modules flanked the tongue module in rostral cortex. LIM1 sits rostral to the buccal pad and LIM2 is lateral to the upper incisor module. The upper incisor module, UIM, is located caudal to LIM1 and lateral to the buccal pad representation. The tongue module, TM, forms the rostralateral border of S1. These four modules were identified in most cases (Fig. 4). No obvious CO-dense modules were identified in S2 or PV. However a CO-dense region in far caudolateral cortex may correspond to auditory or visual cortex, although we have not obtained electrophysiological recordings from this region.

Results of cortical recordings

Electrophysiological recordings were made from five adult naked mole-rats for a total of 1,371 electrode penetration sites in this study. For each recording site, light tactile stimulation was used to identify receptive fields on the body. We found evidence for the existence of three somatosensory areas. S1 was identified as a complete and systematic representation of the contralateral body surface in all five cases. The S1 somatotopic map was orderly and continuous, with only a single representation of each body part. More laterally, S2 was oriented as a mirror image of S1 with a congruent border along midline face and snout. In addition, evidence for a third area, the parietal ventral area (PV), was found in three cases (see Krubitzer et al., 1986).

Figure 5 shows the recording results from case 01-15 in which 460 electrode penetrations were explored (previously published in summary form; Catania and Remple, 2002). The results of these recordings revealed a complete map of the contralateral body and facial surface for S1. Neurons at penetration sites within S1 responded to gentle stimulation of the skin surface, whiskers, teeth, and tongue. The orientation of the body representation in S1 was inverted, with the hindlimb and tail located medially and the face and oral structures located rostralaterally. The physiological map revealed a well-organized somatotopic representation of the body in which small movements of the electrode across the cortical surface resulted in corresponding shifts in the location of receptive fields on the body. This is illustrated in Figure 5 for selected penetrations in three separate progressions of the electrode across the cortex. The progression through the lower body representation (points 1–12) illustrates the orientation of the forelimb and hindlimb in S1, with digits located most rostrally (e.g., penetrations 1 and 12) followed by more proximal aspects of the footpads and subsequently the limbs and thigh regions as the electrode tracked more caudally. Caudal to the limb areas, the ventrum and trunk representations also had discrete receptive fields with the more ventral parts represented more rostrally in cortex and more anterior parts represented more laterally in cortex (see Fig. 5, points 4–6 and A–I). The caudal border of S1 contained tail and lower trunk responses medially and rostral trunk and head laterally (Fig. 5, points A–K).

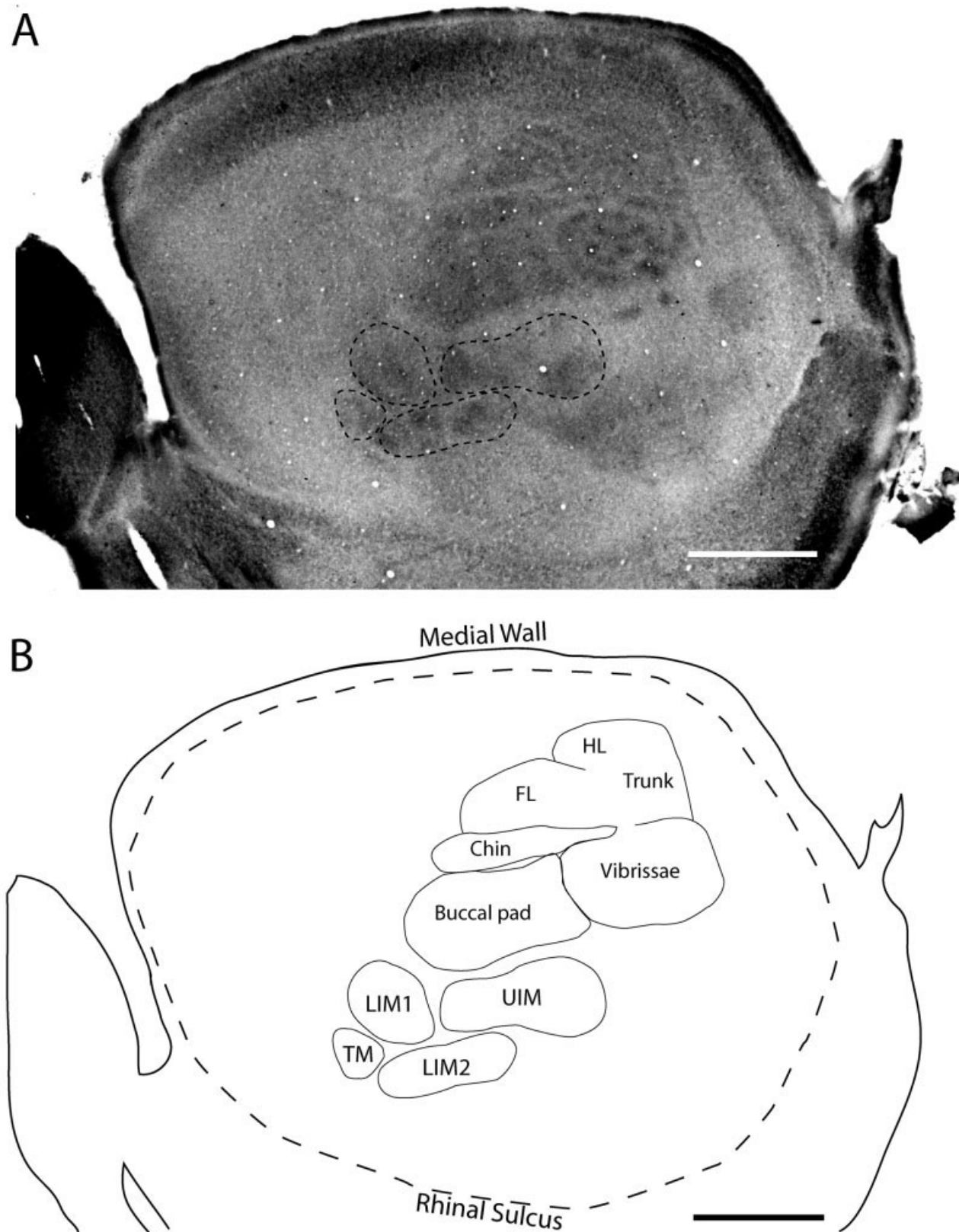


Fig. 3. **A:** A single section of cortical tissue corresponding to layer IV. The section was flattened and cut parallel to the surface before processing for the enzymatic marker for oxidative metabolism, cytochrome oxidase (Wong-Riley et al., 1978; Wong-Riley and Welt, 1980). Cytochrome oxidase processing revealed dense modules correspond-

ing to the predicted locations of S1 body areas and orofacial structures. **B:** Outlines of the cytochrome modules are labeled for clarity. LIM1, lower incisor module 1; LIM2, lower incisor module 2; TM, tongue module; UIM, upper incisor module; HL, hindlimb; FL, forelimb. Scale bars = 1 mm.

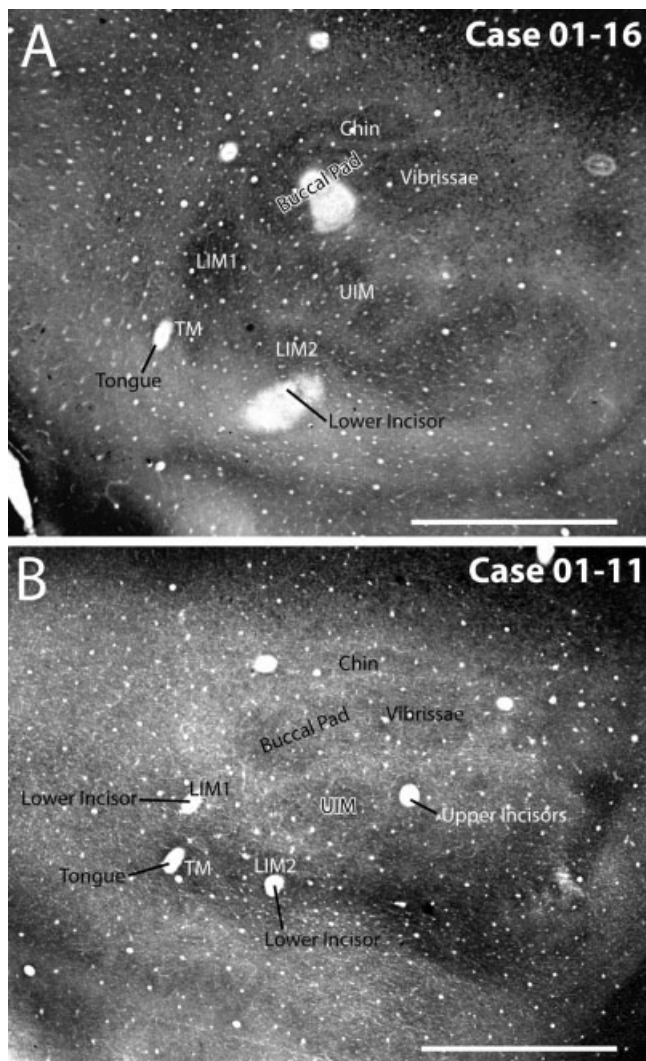


Fig. 4. Cytochrome-dense modules identified in lateral cortex as corresponding to the teeth and tongue. **A,B:** Histology from two physiological cases in which all four lateral modules were distinguishable. Microlesions were placed in these areas during recording sessions to confirm the identification of the modules. LIM1, lower incisor module 1; LIM2, lower incisor module 2; TM, tongue module; UIM, upper incisor module. Scale bars = 1 mm.

We did not detect any unresponsive zones within the S1 representation.

In far lateral S1, the tongue representation formed the rostralateral border of the body representation separated from the buccal pad area by the crescent-shaped lower incisor representation (Fig. 5, penetrations 13–21). Neuronal responses in the lower incisor representation were elicited when any portion of the contralateral incisor was stimulated. The representation of the small whiskers of the buccal pad was located just caudal to the lower incisor area, with most anterior whiskers represented most rostrally in cortex and more posterior whiskers represented most caudally. At its caudal extreme the representation of the buccal pad merged with the representation of the larger facial vibrissae. The representation of the vibrissae

had a spatial organization consistent with the other areas in S1 (Fig. 5, points 15–21). For example, the caudal border of the vibrissae representation responded to stimulation of the rear three accessory whiskers and to midline head where a whisker is situated above the eye (Fig. 5, points 21, H, and I). In the lateral extreme of cortex in the same case there was also a region of mixed incisor responses (Fig. 5, points O and P). We suspect that this region is comparable to the mixed incisor area reported in rats (Remple et al., 2003) and may be the S2 tooth representation.

Similar results were obtained in the other four cases. All of the cases contained consistent representations of the contralateral body surface without any unresponsive zones identified within the S1 area. The forelimb and hindlimb were reliably represented with distal digits most rostral in cortex followed by proximal limb and then the body in more caudal cortex (Figs. 6–9). Cases 01-13 and 01-11 also showed a progression through the forepaw representation with receptive fields moving from D1/D2 to outer digits and palm as the electrode was moved more caudally (Fig. 6, points O, N, and M; Fig. 7, points A–C). The buccal pad and facial vibrissae were also oriented with most anterior whiskers on the face represented rostrally in cortex and more posterior whiskers represented progressively more caudally in the cortex. Cases 01-13, 01-11, and 01-14 provide examples of the progression of receptive fields corresponding to movement of the electrode across the cortical surface for these areas (see Fig. 6, points 6–12; Fig. 7, points G–J and 10–13; and Fig. 9, points 2–8). As might be predicted, the small hairs of the buccal pad located close to the mouth were represented adjacent to the large representations of the incisors in more lateral cortex.

In case 01-13, penetrations 1–3 (Fig. 6) reveal the organization of the chin representation where receptive fields for each of the recording sites moved from ventral to dorsal on the chin as the electrode was moved from rostral to caudal in the cortex. At its caudal extreme, the chin representation merged with the representation of the vibrissae (Fig. 6, penetrations 3–4). The curved portion of this progression line in Figure 6 (points 4–8) reveals the topography of the vibrissae representation in S1, and the continuity between the representation of vibrissae and the facial whiskers on the chin and buccal region.

The representation of the snout was identified in S1 of each case at the lateral border of the vibrissae in caudal cortex. Neuronal responses in this area could be elicited by stimulating the skin and small hairs of the midline nose and nostril region (see Fig. 8, penetration 11; Fig. 5, penetration K). The snout representation was significant in marking the border between areas S1 and S2. The location of S2 could be distinguished from S1 by its lower CO density allowing for delineation of the two areas by histological means (Fig. 3).

A striking feature of somatosensory cortex in mole-rats was the large representation of the upper and lower incisors. In all of the cases there was a large crescent-shaped area in rostralateral cortex that was responsive to gentle stimulation of the contralateral lower incisor (Figs. 5–9). The lower incisors are independently mobile and can be separated to restrict stimulation to a single tooth. Care was taken to confirm that the receptive fields for the lower incisor were exclusively contralateral. We found only a single bilateral receptive field for the lower incisors lo-

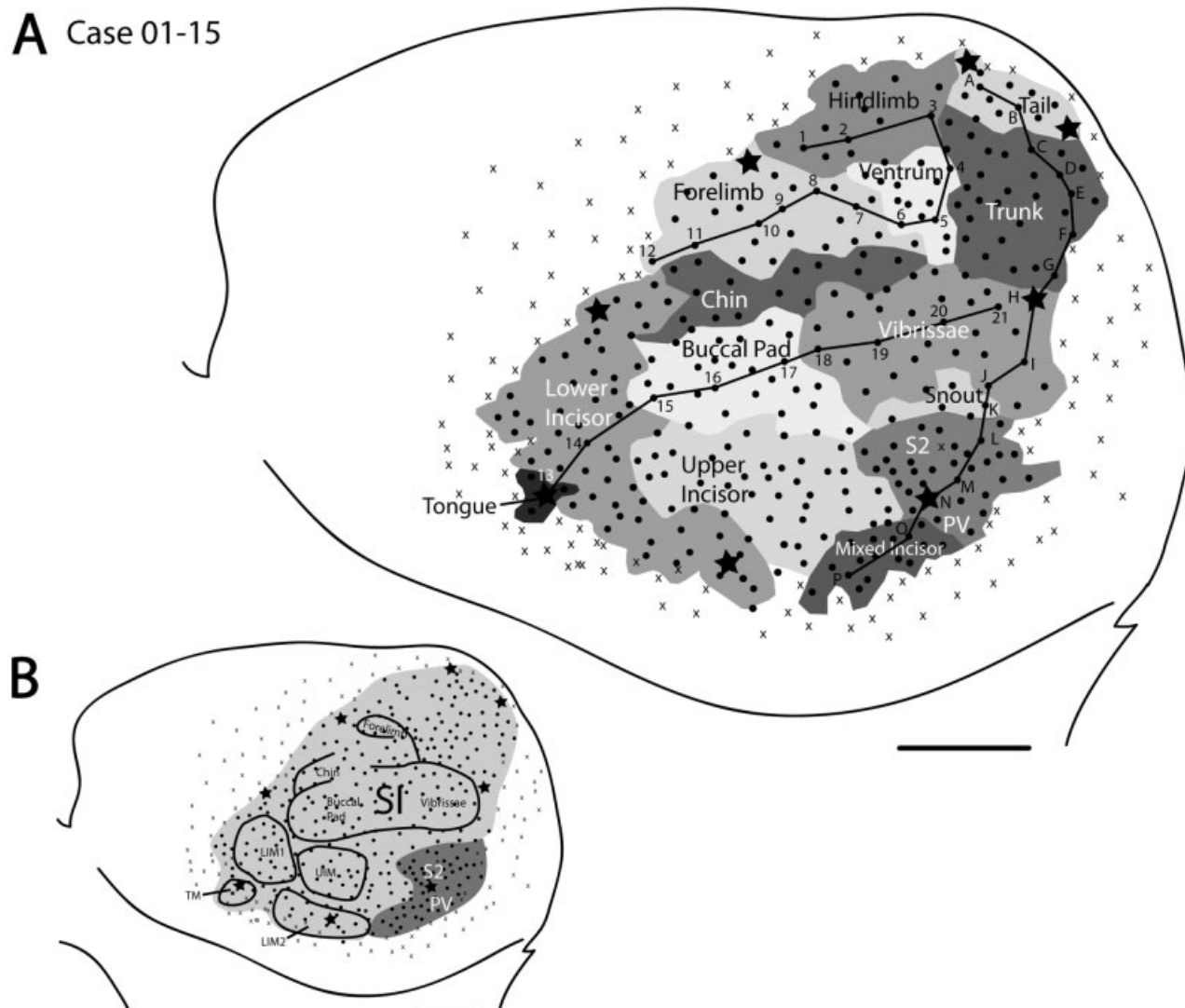
A Case 01-15

Fig. 5. The somatotopy and chemoarchitecture of S1. **A:** Physiologically derived map of somatosensory cortex. Gray shaded regions identify the extent of each body and orofacial area. Dots indicate penetration sites, stars represent microlesions, and Xs mark sites that did not elicit a response. **B:** Outlines of cytochrome-dense modules aligned with recording penetrations using the microlesions to

determine position. **C:** Receptive field progressions drawn on schematic diagrams of naked mole-rat body regions. LIM1, lower incisor module 1; LIM2, lower incisor module 2; TM, tongue module; UIM, upper incisor module; HL, hindlimb; FL, forelimb. Scale bars = 1 mm in A, B.

cated in the PV area of case 01-14 (Fig. 12). It should also be noted that at different penetration sites within the lower incisor representation, maximal neuronal responses were achieved using different types of stimulation. Some areas were preferentially stimulated by stroking the enamel surface, while others responded best to gentle tapping. There did not appear to be an obvious organization or regionalization of neurons with a particular stimulation preference; however, this was not thoroughly explored. Similar regional differences in stimulation preference were noted for the upper incisor representation.

The upper incisor representation was located below the buccal pad area in lateral cortex. Neurons in this region consistently responded to stimulation of either upper in-

cisor. Because the two upper incisors are always in contact with one another, we could not determine whether responses were bilateral or simply the result of simultaneous stimulation of both incisors. The caudal border of the upper incisor area was at times hard to differentiate physiologically from the mixed incisor area that corresponds to the secondary somatosensory areas (Figs. 6, 9). The cytochrome-dense oral module for the upper incisors (UIM) was the best determinant for the boundary of the area.

Topography of S2

At the posterior and lateral border of S1, we identified the secondary somatosensory area (S2) as an additional representation of the contralateral body surface. This rep-

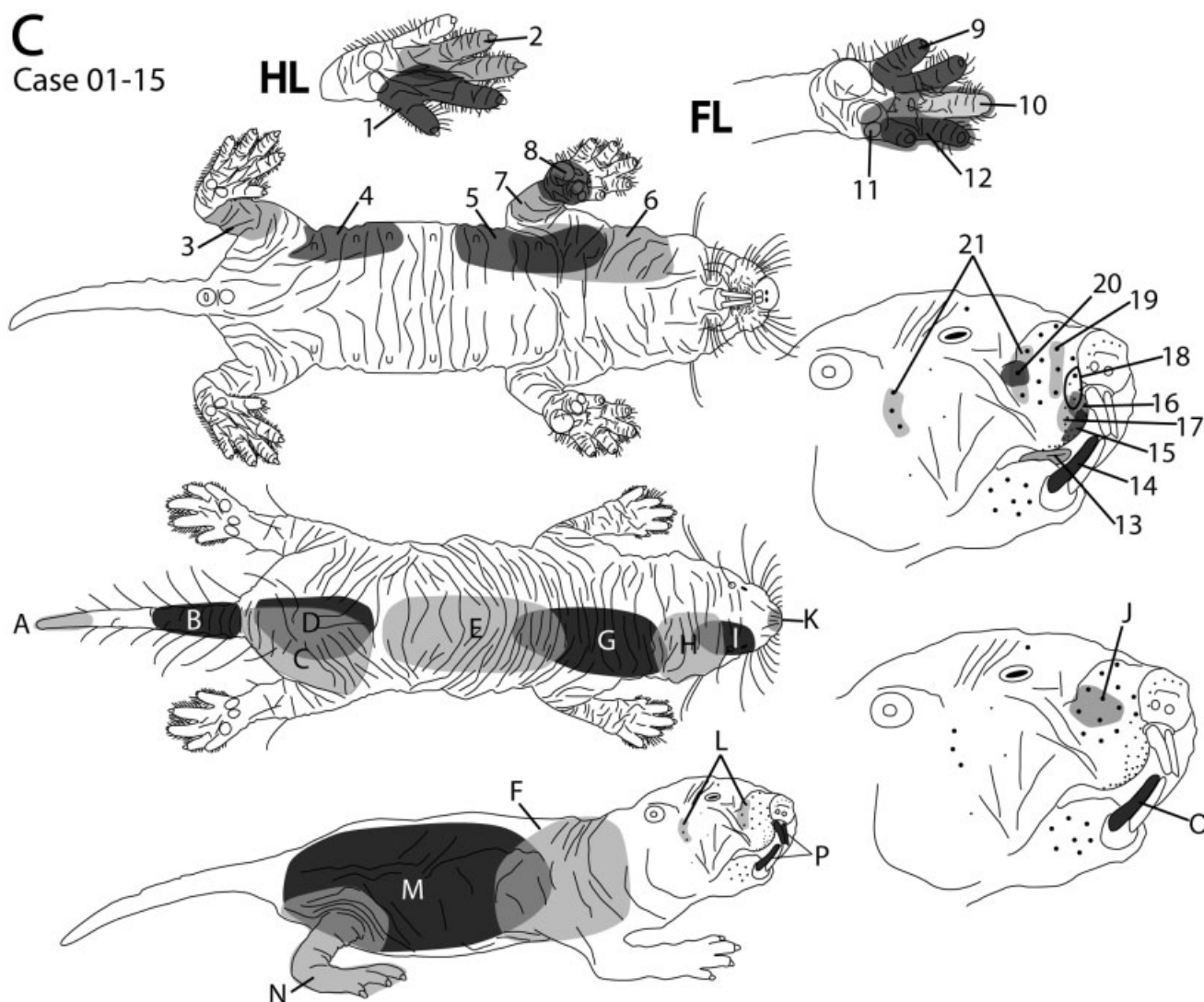


Figure 5 (Continued)

resentation was a smaller, mirror-image of S1 somatotopy, with the common border located at the midline of the face and snout areas. The body representation in S2 had an upright orientation, with the face located medially and the limbs and trunk located laterally. Only three of the experimental cases (Figs. 10–12) provided enough detail in that region to reveal the topography of the representation. In the other two cases we were not able to explore the full lateral extent of this region, but electrode penetrations in each case did suggest the presence of S2 (Fig. 7, point 15; Fig. 8, points 12 and 13).

Larger receptive fields were a common characteristic observed in S2. In case 01-15, neurons at one penetration site responded to stimulation across the entire trunk region or the combined hindlimb and shoulder territory (Fig. 10, points C and D). The receptive fields corresponding to the vibrissae were often expanded in S2 as compared to similar sites in S1 (Fig. 7, compare points 12 and 13 with point 15). We were able to calculate the receptive field areas for all of the S2 penetration sites collected over the

five cases and compare these data to a population of receptive fields from the recordings in S1 (Table 1). All of the body and face areas in S2 had larger receptive fields as compared to the equivalent regions in S1.

Despite these relatively large receptive fields, the overall topography of the body representation could still be determined (Fig. 10, 1–6 and A–F). The face area was oriented in cortex with anterior sensory surfaces (teeth and rostral buccal pad) represented in more rostral cortex and posterior sensory surfaces (posterior vibrissae on the cheek) represented in more caudal cortex. The forelimb and trunk representations were found lateral to the face representation (Fig. 10, points A–C). The hindlimb representation in S2 was located lateral to the forelimb, just anterior to the representation of the trunk (Fig. 10, points F, E, and D).

In case 01-13 the somatotopy for the S2 face representation was also determined along the rostral-caudal axis of cortex (Fig. 11). Although receptive fields were larger than those in S1, the same sequential nature of responses was

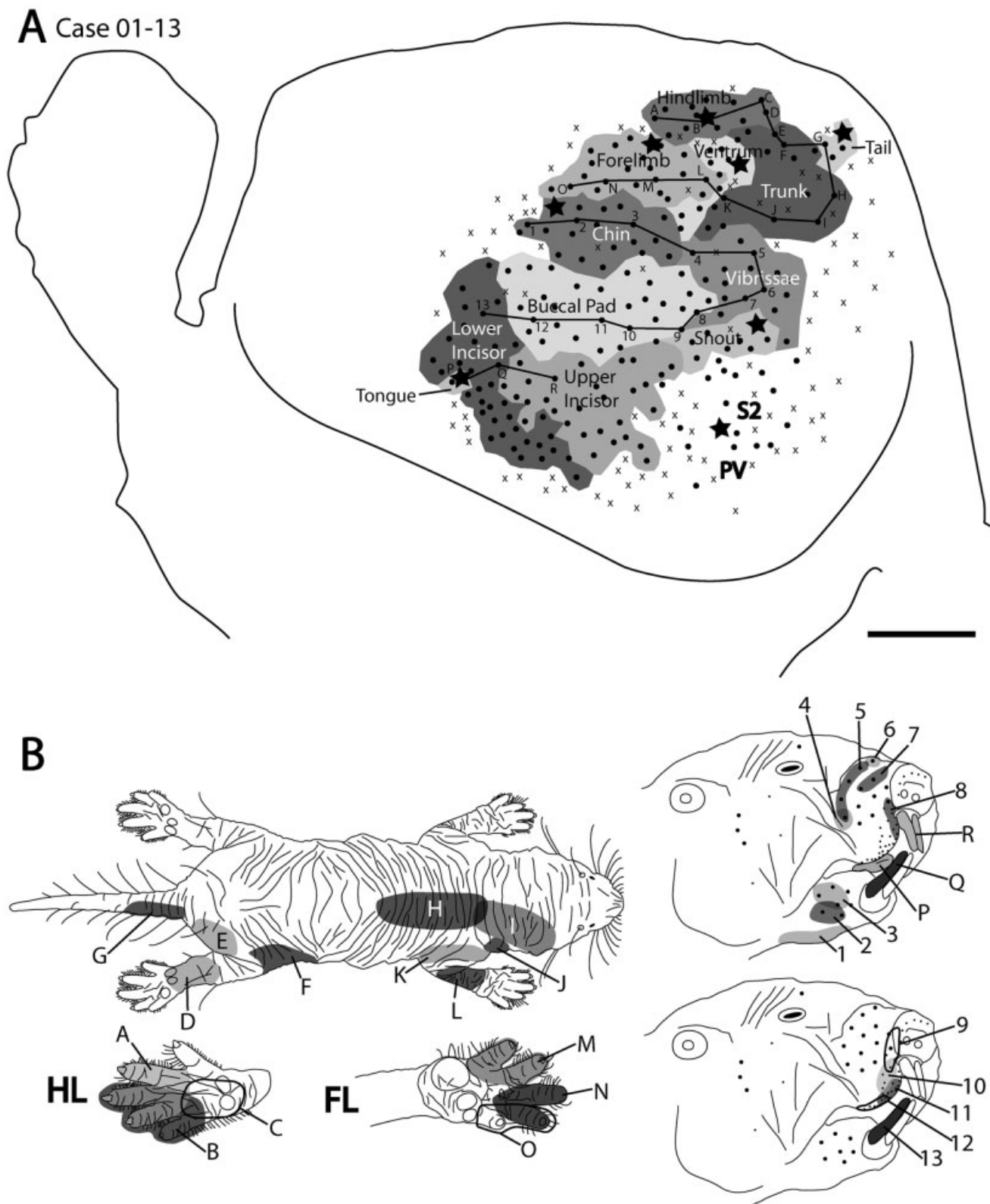


Fig. 6. The somatotopy and chemoarchitecture of S1. **A:** Physiologically derived map of somatosensory cortex. Gray shaded regions identify the extent of each body and orofacial area. Dots indicate penetration sites, stars represent microlesions, and Xs mark sites

that did not elicit a response. **B:** Receptive field progressions drawn on schematic diagrams of naked mole-rat body regions. HL, hindlimb; FL, forelimb. Scale bar = 1 mm.

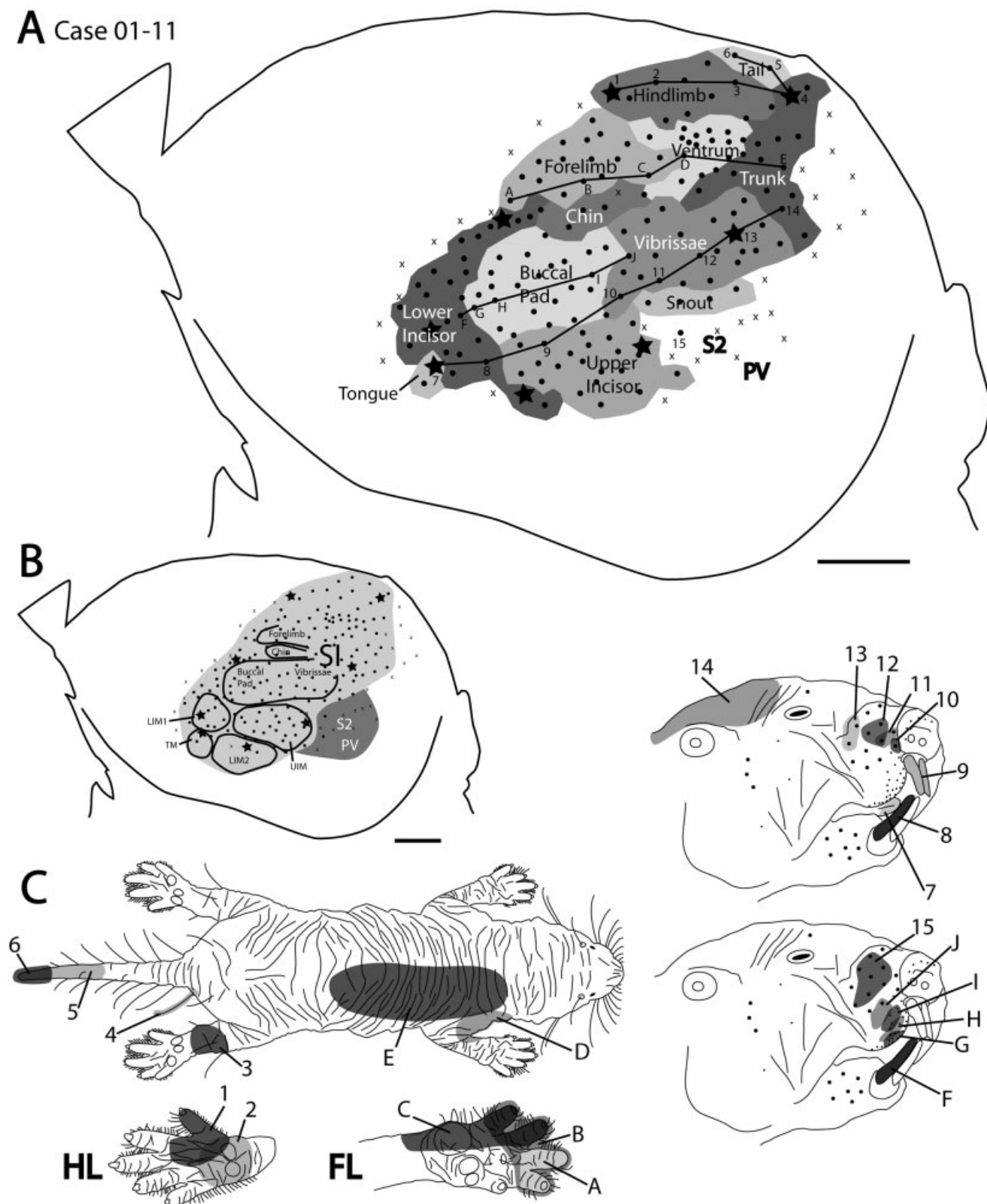


Fig. 7. The somatotopy and chemoarchitecture of S1. **A:** Physiologically derived map of somatosensory cortex. Gray shaded regions identify the extent of each body and orofacial area. Dots indicate penetration sites, stars represent microlesions, and Xs mark sites that did not elicit a response. **B:** Outlines of cytochrome-dense modules aligned with recording penetrations using the microlesions to

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A Case 01-16

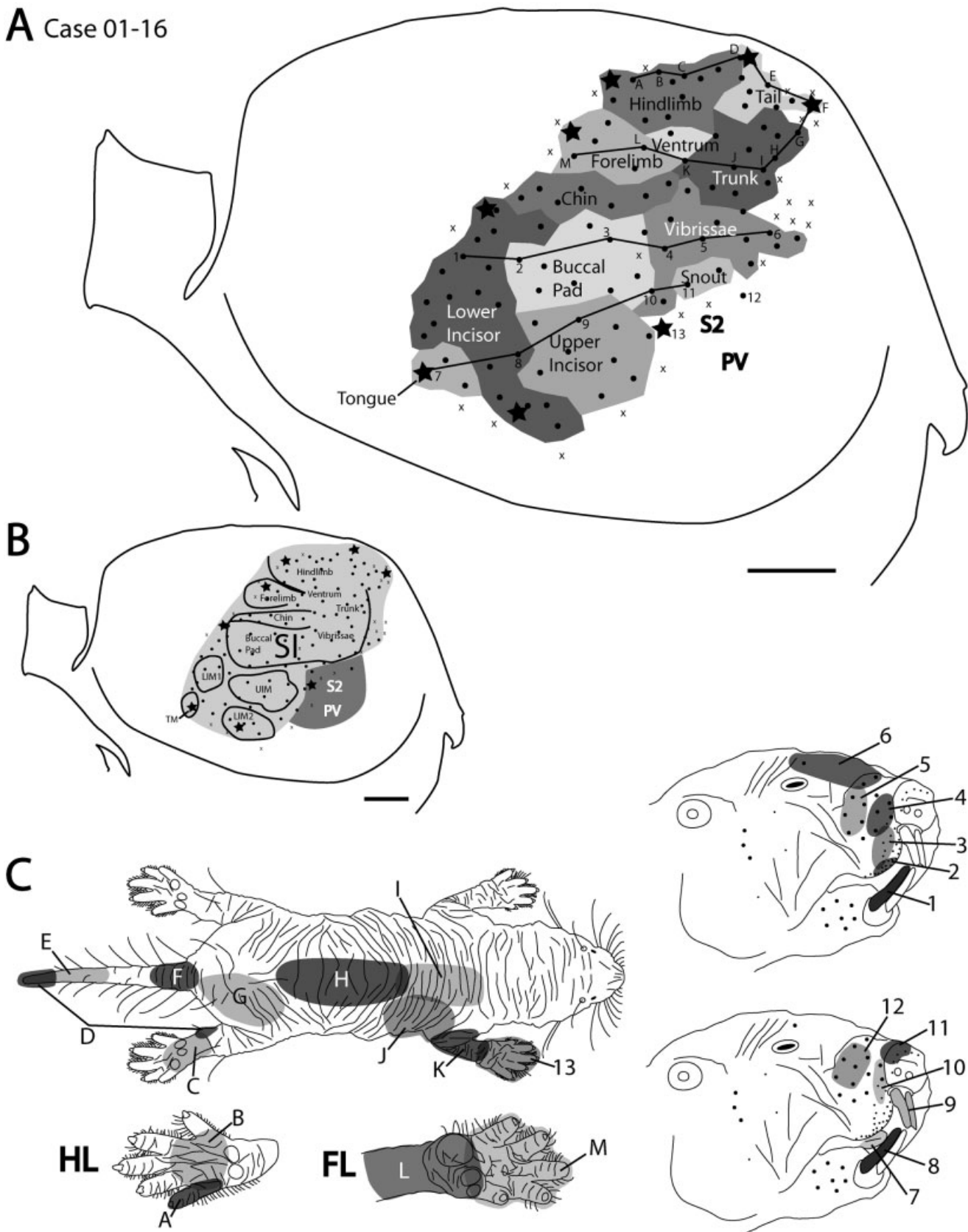


Fig. 8. The somatotopy and chemoarchitecture of S1. **A:** Physiologically derived map of somatosensory cortex. Gray shaded regions identify the extent of each body and orofacial area. Dots indicate penetration sites, stars represent microlesions, and Xs mark sites that did not elicit a response. **B:** Outlines of cytochrome-dense modules aligned with recording penetrations using the microlesions to

determine position. **C:** Receptive field progressions drawn on schematic diagrams of naked mole-rat body regions. LIM1, lower incisor module 1; LIM2, lower incisor module 2; TM, tongue module; UIM, upper incisor module; HL, hindlimb; FL, forelimb. Scale bars = 1 mm in A, B.

A Case 01-14

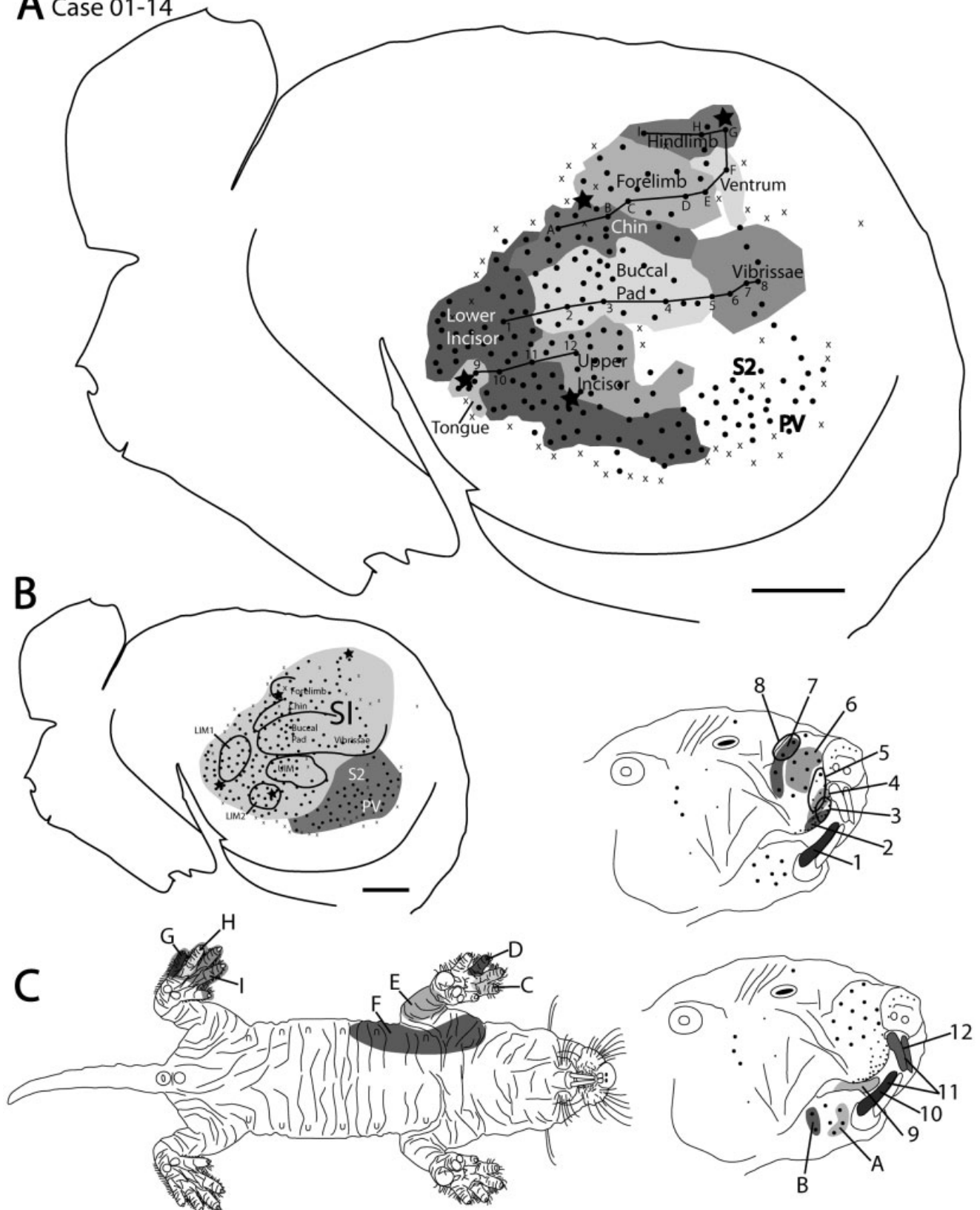


Fig. 9. The somatotopy and cytoarchitecture of S1. **A:** Physiologically derived map of somatosensory cortex. Gray shaded regions identify the extent of each body and orofacial area. Dots indicate penetration sites, stars represent microlesions, and Xs mark sites that did not elicit a response. **B:** Outlines of cytochrome-dense modules aligned

with recording penetrations using the microlesions to determine position. **C:** Receptive field progressions drawn on schematic diagrams of naked mole-rat body regions. LIM1, lower incisor module 1; LIM2, lower incisor module 2; TM, tongue module; UIM, upper incisor module. Scale bar = 1 mm in A, B.

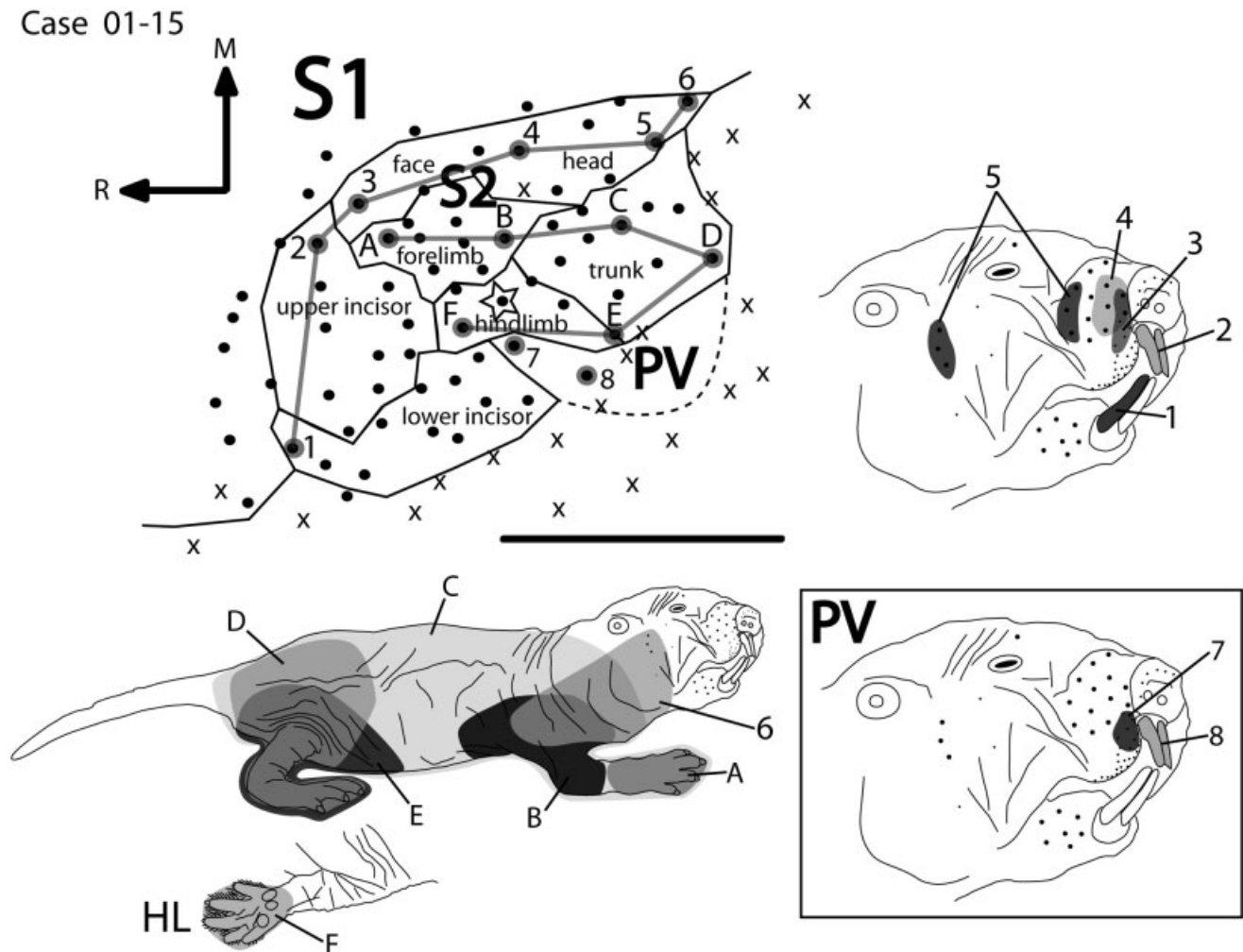


Fig. 10. The somatotopic organization of lateral cortex. Physiologically derived map of lateral cortex revealing the extent of area S2 and presence of area PV. Dots indicate penetration sites, stars represent microlesions, and Xs mark sites that did not elicit a response. Recep-

tive field progressions drawn on schematic diagrams of naked mole-rat body regions, boxed area designates responses from PV area. The medial (M) and rostral (R) directions are indicated in the upper left-hand corner. Scale bar = 1 mm.

found beginning at the rostral-most S2 sites, which contained neurons responsive to the incisors followed by buccal pad and then vibrissae as the electrode was moved more caudally (Fig. 11, penetrations 1–5). In this case there were few exclusively trunk responses, but receptive fields progressed from exclusively limbs in rostral cortex (e.g., Fig. 11 penetrations A and G) to include large portions of the limbs and body in more caudal cortex (penetrations D and C). In case 01-14, a similar overall organization was found for S2. Figure 12 illustrates the electrode penetrations and selected receptive fields for this case.

Third somatosensory representation

In a number of cases, recording sites located lateral to the S2 area responded to stimulation of the contralateral body surface in a pattern that suggested the presence of a third somatosensory area (Figs. 10–12). This third area is likely the parietal ventral area (PV), an area that has been identified in a variety of other mammals including small-

brained rodent species such as the rat and squirrel (Krubitzer et al., 1986; Koralek et al., 1990; Fabri and Burton, 1991; Krubitzer and Calford, 1992; Slutsky et al., 2000; Remple et al., 2003). The area was most thoroughly explored in case 01-14, where a number of neurons responded to stimulation of the face and rostral trunk (Fig. 12, points 6–11). This lateral representation of the face suggests that PV has an inverted orientation with the limbs and trunk forming the border between S2 and PV.

In case 01-15 a recording site responsive to the face was separated from the S2 face representation by the representations of the forelimb, hindlimb, and lower trunk areas (Fig. 10, point 7). There was also a site responsive to the upper incisors in this region (Fig. 10, point 8). In case 01-13 a single penetration in far lateral cortex responded to stimulation of the anterior shoulder (Fig. 11, point H). This site was separated from the S2 forelimb representation by the S2 hindlimb area, suggesting that it was part of another representation. The receptive fields within this

Case 01-13

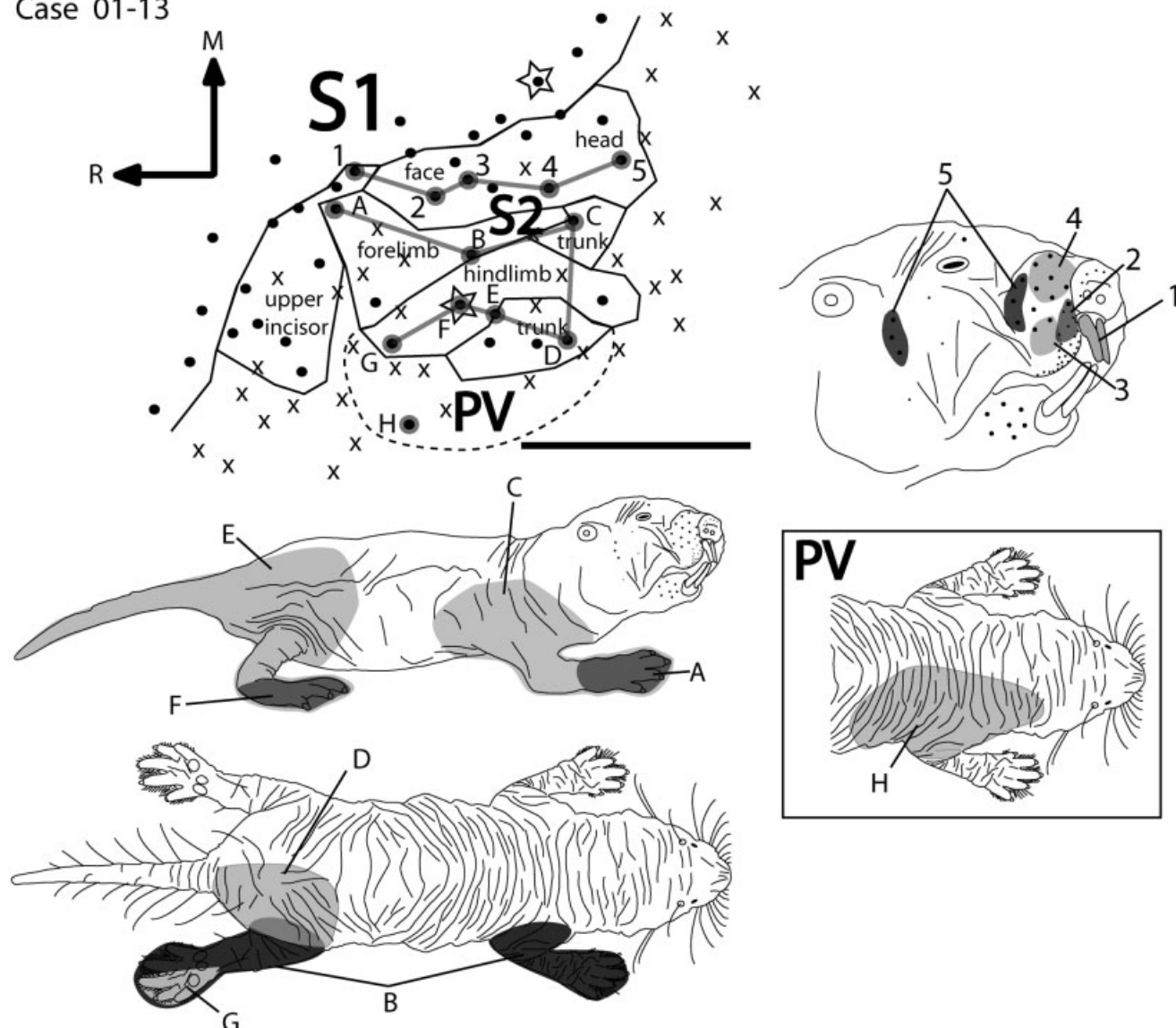


Fig. 11. The somatotopic organization of lateral cortex. Physiologically derived map of lateral cortex revealing the extent of area S2 and presence of area PV. Dots indicate penetration sites, stars represent microlesions, and Xs mark sites that did not elicit a response. Recep-

tive field progressions are drawn on schematic diagrams of naked mole-rat body regions, boxed area designates responses from PV area. The medial (M) and rostral (R) directions are indicated in the upper left-hand corner. Scale bar = 1 mm.

lateral territory were also larger than those found in S1 or S2, particularly for the penetration sites responsive to the face. These results are consistent with findings in a number of other small mammals suggesting that cortex lateral to S1 contains both S2 and PV.

DISCUSSION

The present results suggest that the somatosensory cortex of the naked mole-rat contains at least two discrete areas: S1 and S2, and likely a third, which we hypothesize to be PV, the parietal ventral area. This hypothesis is based on similarities in location, orientation, and receptive field size to PV in other small-brained mammals. A

schematic representation of this topographical organization as determined by our physiological recordings is diagrammed in Figure 13. This work extends our previous brief report of cortical organization in the naked mole-rat (Catania and Remple, 2002) by identifying additional somatosensory areas in lateral cortex, showing the details of receptive fields associated with each area, and revealing a pattern of cytochrome oxidase modules and barrels that corresponds to the representations of various body parts in S1.

The most obvious characteristic of brain organization in this species is the domination of the neocortex by somatosensory inputs. Nearly all of the caudal neocortex is taken up by somatosensation. This expansion of somatosensory

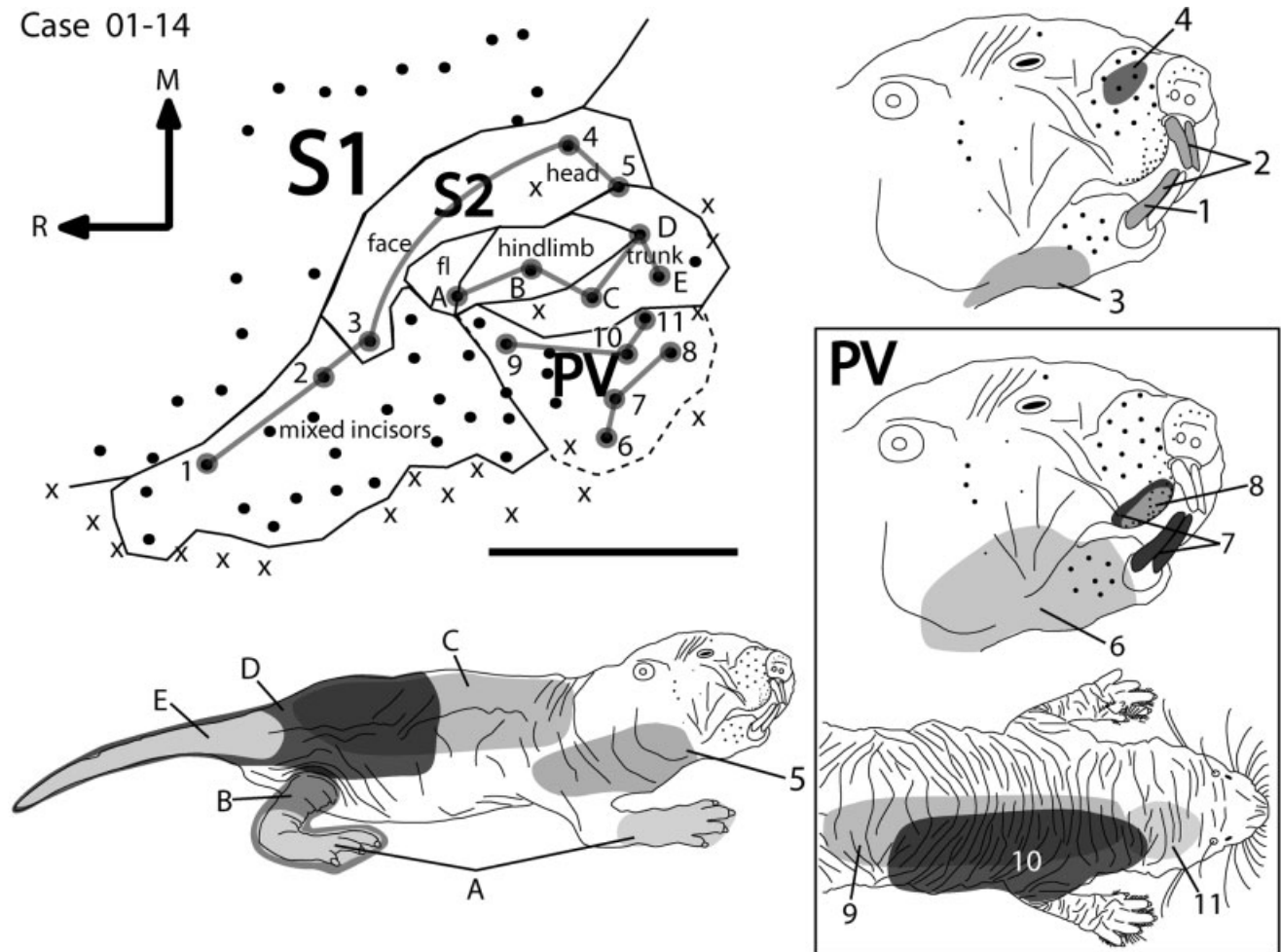


Fig. 12. The somatotopic organization of lateral cortex. Physiologically derived map of lateral cortex revealing the extent of area S2 and presence of area PV. Dots indicate penetration sites, stars represent microlesions, Xs mark sites that did not elicit a response, and fl =

forelimb. Receptive field progressions are drawn on schematic diagrams of naked mole-rat body regions, boxed area designates responses from PV area. The medial (M) and rostral (R) directions are indicated in the upper left-hand corner. Scale bar = 1 mm.

cortex at the expense of other cortical areas is a testament to the importance of touch in the mole-rat's fossorial environment and illustrates the flexibility of cortical organization possible in the face of selection to enhance processing of a particular modality.

S1 in the naked mole-rat

S1 has been identified in every mammalian species that has been investigated and is characterized by a complete somatotopic map of the contralateral body surface (Woolsey and Fairman, 1946; Woolsey, 1958; Kaas, 1983). In primates, area 3b is considered the homolog to S1, based on its many similarities to S1 in a wide range of mammalian species (Kaas, 1983). Multiple criteria are generally used to identify cortical areas, as many subdivisions are difficult to distinguish. Although S1 is one of the more distinctive cortical areas, parts of this region may be difficult to identify in specialized species. For example, in moles the S2 forelimb representation is larger and more distinct than the S1 forelimb representation. This led to the misidentification of the S2 forelimb representation as

part of S1 in moles (Catania, 2000). Similarly, in bats the S1 wing representation is in an unexpected location, with the distal aspect of the modified forelimb displaced to the caudal portion of S1, whereas it is located in the rostral portion of S1 in other mammals (Calford et al., 1985; Krubitzer and Calford, 1992).

In mole-rats, S1 can be clearly identified based on its topographic representation of the contralateral body surface and its relative location, orientation, histological appearance, and response properties. However S1 in naked mole-rats is unusual in a number of ways. For example, it is considerably larger, as a proportion of the neocortex, than S1 of other rodents that have been investigated and it lacks an unresponsive zone within the territory. Comparisons with laboratory rat cortex (Catania and Remple, 2002) indicate that S1 in mole-rats has increased in size by ~50% as a percentage of total neocortex (assuming the ancestral condition was similar to laboratory rats). A similar but less extreme expansion of S1 has been reported for the blind mole-rat (Necker et al., 1992). Another unusual characteristic of S1 in naked mole-rats, also reported for

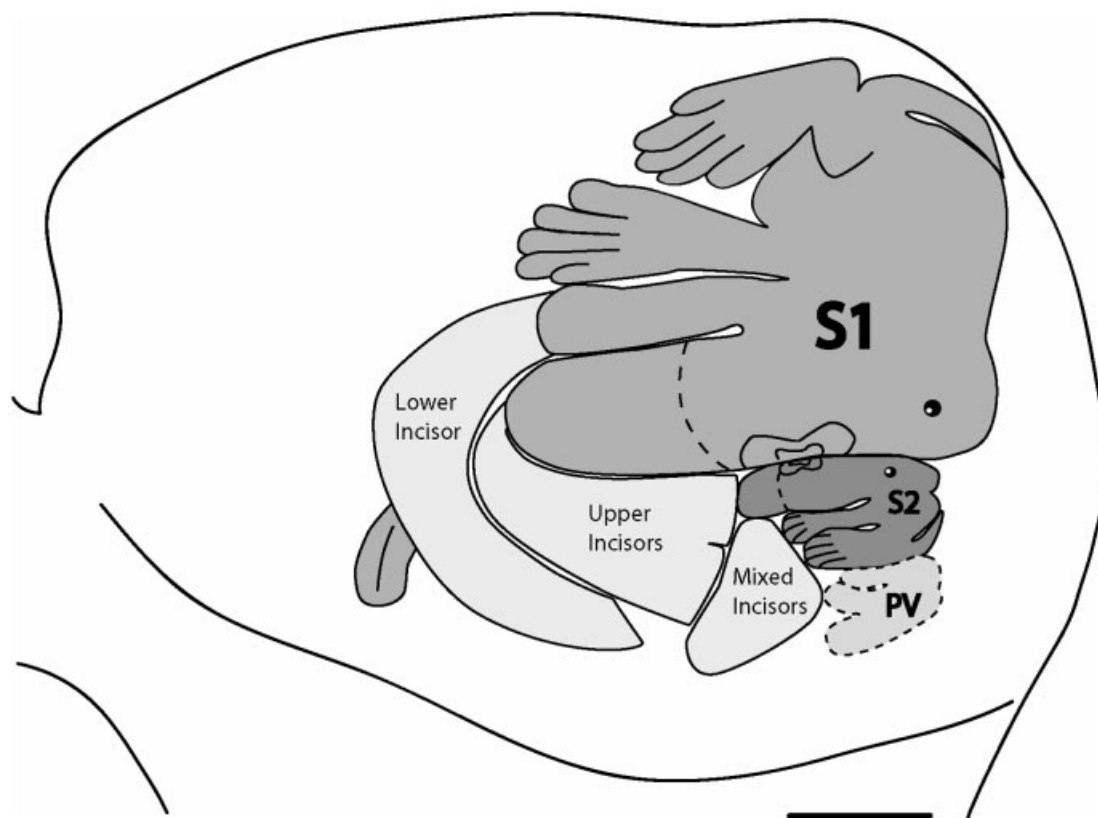


Fig. 13. Summary schematic of the somatotopic organization, location, and relative size of areas: S1, S2, and a region presumed to be the parietal ventral area (PV), dotted line. The large representations of the upper and lower incisors and the rostral tongue area are all part of S1. The mixed incisor region is composed of neurons responsive to the upper incisor, lower incisor, or both.

blind mole-rats, is its extension into far medial and caudal cortex into areas usually occupied by primary visual cortex (V1) (Necker et al., 1992; Catania and Remple, 2002). It appears that much of the area normally devoted to vision in mammals is devoted to somatosensation in mole-rats generally. Finally, the most distinctive specialization in naked mole-rat S1 is the allocation of a huge territory of lateral cortex to the representation of the dentition. The representation of these behaviorally important upper and lower incisors occupies over 30% of S1 (Catania and Remple, 2002). This unprecedented cortical magnification of the incisors in S1 is greater than magnification of the posterior medial barrel subfield in somatosensory cortex of laboratory rats (estimated from Chapin et al., 1987). Behavioral observations show that mole-rats rely heavily on their incisors for foraging, digging, and moving objects (Lacey et al., 1991). This behavioral dependence is certainly a major contributor to the cortical magnification of the dentition in cortex. The large representation of the teeth has made mole-rats a useful species for investigating the plasticity of cortical areas representing the dentition (Henry et al., 2005). The cortical areas are large enough to allow detection of changes that might otherwise go unobserved in brains with smaller representations of the teeth.

During our receptive field mapping we observed that neurons in the dentition representation appeared to have

stimulation preferences for achieving a maximal response. Some penetration sites were most responsive when the incisor was lightly tapped with a soft probe, while for others, brushing the probe down the incisor's entire length elicited the greatest response. Although we were not able to discern the organization of this stimulation preference in cortex, it is possible that there is a cortical compartmentalization of mechanoreceptor input from the periodontal ligament in the S1 dentition representation. The mechanoreceptors in the periodontal ligament of the mole-rat have not yet been explored in detail, but in rats the periodontal ligament is largely composed of unencapsulated ruffini-like corpuscles, which have been demonstrated to have directional sensitivity (Chambers et al., 1972; Sato et al., 1988; Byers and Dong, 1989; Takahashi-Iwanaga et al., 1997). It is likely that similar mechanoreceptors reside in the naked mole-rat periodontal ligament and that the stimulation preference we observed was due to the directional specificity of these mechanoreceptors in the periphery. Further investigation into the population of mechanoreceptors in the periodontal ligament and their physiological characteristics is needed to confidently relate mechanoreceptor stimulus preference to the cortical representation.

Few previous studies have included investigation of the dentition representation in cortex. This is likely due to the far lateral location of these representations in cortex,

which makes them hard to access surgically, combined with the limiting nature of most stereotaxic equipment that typically have a bar located in front of the incisors, making stimulation of the oral areas difficult. However, in nonhuman primates a number of investigators have explored lateral 3b and have consistently identified large representations of the oral structures, including areas related to both ipsilateral and contralateral teeth (Manger et al., 1996, 1995; Jain et al., 2001). Few studies have examined the dentition representation in small mammals, although Carvell and Simons (1986) reported six penetration sites responsive to the incisors in the mouse. However these responses were located in far caudolateral cortex and attributed to S2, rather than S1.

More recently, the S1 dentition representation was investigated in rats, providing a more detailed account of the area. Remple et al. (2003) found that ~7% of S1 corresponded to the contralateral incisors in the rat. This can be compared to mole-rat dentition representation, which occupies ~30% of S1 (Catania and Remple, 2002)—a four-fold difference in the percent of S1 devoted to the incisors in these two closely related species. An obvious explanation for this difference is the greater dependence on information from the dentition in mole-rats—which are functionally blind. However the relative contributions of behavioral versus anatomical (e.g., innervation density) influences on the development of the large incisor representations have not been investigated in mole-rats (but see Henry et al., 2005, for plasticity of this region in response to tooth loss).

In mole-rats, the buccal pad, chin, and vibrissae representations were also more greatly magnified in cortex. Together they occupied over one-third of S1 territory, equivalent to the representation of the entire lower body from the neck to the tail (Catania and Remple, 2002). The receptive fields for the neurons in the facial representations were substantially smaller than those recorded in the lower body areas (Table 1). A number of early investigations suggested that the differential magnification of sensory surfaces in cortex is due to differences in peripheral innervation density (Whitteridge, 1963; Lee and Woolsey, 1975; Malpelli and Baker, 1975). However, more recent studies in the visual (Azzopardi and Cowey, 1993) and somatosensory (Catania and Kaas, 1997) systems suggest cortical magnification may also reflect the behavioral importance of peripheral sensory structures, rather than simply innervation density. The teeth and whiskers are invaluable to the mole-rat when navigating through their underground tunnels and foraging for food (Jarvis, 1991). However, the relative role of innervation density and behavioral use of the dentition remains to be explored in mole-rats.

Cytochrome oxidase-dense regions of S1

Processing flattened cortical tissue for cytochrome oxidase histochemistry has been used to characterize rat and mouse sensory cortex (Land, 1985; Wallace, 1987), and is particularly effective for distinguishing layer IV granule cell aggregates in primary somatosensory areas. Using this technique, we identified histological correlates of the S1 body representation and orofacial structures in the naked mole-rat. The histologically visible subdivisions in naked mole-rat cortex are not as distinct as those in S1 of laboratory rats and mice. However, we were able to identify a number of subdivisions that correspond to represen-

tations of various body parts. Indeed, investigators experienced at examining the histology of cortex in laboratory rats and mice will probably recognize familiar subdivisions representing the various body parts in flattened cortex (Fig. 3). Within the vibrissae area, a number of individual cytochrome oxidase-dense modules were visible. These barrel-like submodules in the vibrissae area are almost certainly related to the individual whiskers on the mole-rat's face, as has been reported for other rodents (Welker and Woolsey, 1974; Woolsey et al., 1975). The mole-rat S1 barrel pattern is similar to the layout of whiskers on the mole-rats face (Crish et al., 2003); however, the barrels in the mole-rat cortex are less discrete and do not always have clear septal borders between one another (Fig. 3). This may be due to the smaller size, and perhaps lower innervation density, of sensory hairs on the face of mole-rats compared to laboratory rats and mice. For example, Welker and Van der Loos (1986) found that mice bred for supernumerary whiskers only had histologically visible barrels represented in cortex for those whiskers with a threshold number of afferents. It may be that mole-rat facial whiskers are close to this threshold, and that the sensory hairs of the buccal pad (which had no obvious barrel representation in mole-rat cortex) are below this threshold.

The S1 chin area of the mole rat was represented directly medial to the buccal pad and rostral vibrissae area. There was a narrow, perigranular, intercalated zone between the chin and buccal pad areas that allowed for visual identification of the borders of the two regions (Fig. 3). There also appeared to be a second perigranular strip along the medial border of the chin representation, separating it from the forelimb representation. This is similar to what is seen at the rat chin/forelimb border, although the perigranular area is less distinct in the mole-rat (Chapin and Lin, 1984; Chapin et al., 1987). The forelimb, hindlimb, and trunk representations all appeared histologically continuous, with uniformly dense cytochrome oxidase activity. This uniformity is unlike the S1 lower body representations in the rat, where submodules for each of the forepaw digits are visible (Dawson and Killackey, 1987). The rostral and caudal borders of the body were delineated by a drop in cytochrome oxidase activity characteristic of the surrounding motor and sensory territories.

The lateral areas in S1, corresponding to the incisors and tongue representations, contained visible modules that corresponded to these sensory representations. Two cytochrome oxidase-dense modules represented the lower incisor (LIM1 and LIM2), one represented the upper incisor (UIM), and one represented the tongue (TM) (Figs. 3, 4). Similar cytochrome oxidase-dark modules have been identified in the lateral cortex of rats; however, rat S1 appears to contain only one module related to the lower incisor and one related to the tongue (Remple et al., 2003). There was no obvious module corresponding to the representation of the upper incisor in rats. However, the proximity of the upper incisor representation to the buccal pad representation in rat S1 could make identification of such a module difficult. Alternatively, the upper incisor may play a less important sensory role in laboratory rats and have a lower innervation density that is below a threshold for visible modular representations in the cortex (e.g., Welker and Van der Loos, 1986). Modular representations of the dentition have been identified in S1 of squirrel

monkeys and owl monkeys (Jain et al., 2001). It seems likely that similar modules representing dentition may be present in the cortex of other mammals, but few studies have examined these areas in the far lateral cortex.

We were not able to distinguish clear histological correlates of S2 and presumptive PV in the mole-rat. In general, secondary areas have a less distinct histological appearance, as they receive fewer direct thalamocortical projections and have a sparser granule cell layer than primary areas. There were some regions of high cytochrome oxidase-density in far caudolateral cortex, but this area did not contain neurons that responded to tactile stimulation of the body. It seems possible that this area corresponds to auditory cortex (see Necker, 1992) but we did not explore this possibility.

S2 and presumptive PV

An important goal of our investigations of brain organization in mammals is to determine the number and identity of cortical areas. This information is required for understanding how individual species are specialized for processing sensory information and for identifying trends in mammalian brain evolution. Somatosensory areas in lateral cortex have been historically difficult to explore because of their relatively small size, correspondingly large receptive fields, and locations in areas of the brain that are difficult to access in some species. However, we were able to explore these areas in several mole-rat cases and found evidence for the existence of both S2 and a third somatosensory area, which appears to be the parietal ventral area (PV).

Evidence for these areas comes from the electrode penetrations that suggest there are two representations of the contralateral body surface in cortex lateral to S1. The more medial of these two fields, S2 appeared to contain a complete representation of the body with a common border with S1 along the midline snout and vibrissae representations. The limbs and trunk were represented more laterally in cortex, such that S2 was a mirror image of S1. S2 contained receptive fields that were larger than the receptive fields identified in S1 (Figs. 5–12; Table 1) as is commonly reported for S2 (Nelson et al., 1979; Carvell and Simons, 1986; Remple et al., 2003) and is consistent with the more general observation that receptive field size on the body is inversely proportional to representational area in the cortex (Sur et al., 1980).

We also found evidence for a third somatosensory area, which resembles the PV described in a range of other small mammals (see Krubitzer et al., 1995b). In particular, this area is similar in its position (lateral to S2) to PV as described in closely related laboratory rats (Fabri and Burton, 1991; Remple et al., 2003). In addition to PV, Fabri and Burton (1991) also identified a more lateral somatosensory area the parietal rhinal area (PR) based on connections with S1. If PR exists in mole-rats, is likely to be just lateral to our presumptive PV, as PR in rats is directly adjacent to the rhinal sulcus and does not share a common border with S2. PR in rats has also been associated with visceral sensory processing, making it an unlikely candidate for the area we identified in mole-rats that contained neurons responsive to cutaneous mechanoreceptors of the skin (Cechetto and Saper, 1987; Fabri and Burton, 1991).

PV has been commonly identified in a number of small-brained mammals including the squirrel and rat (Kru-

bitzer et al., 1986; Fabri and Burton, 1991; Remple et al., 2003). However, these two species appear to have different arrangements of the PV body representation. In squirrels PV is located rostralateral to S2. In contrast, PV in rats has been shown to reside lateral to S2 in an inverted orientation, completely mirroring the body and face arrangement of the more medial S2 (Fabri and Burton, 1991; Remple et al., 2003). In this regard, the organization we report for S2 and PV in mole-rats appears to be similar to that seen in rats.

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